Where $m_1, m_2, m_3 \ldots\ldots$ are magnification of individual lenses.

EXAMPLE 6.16

What is the focal length of the combination if the lenses of focal lengths –70 cm and 150 cm are in contact? What is the power of the combination?

Solution

Given, focal length of first lens, $f_1 = -70$ cm, focal length of second lens, $f_2 = 150$ cm.

Equation for focal length of lenses in contact, $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$

Substituting the values,

$$\frac{1}{f} = \frac{1}{-70} + \frac{1}{150} = -\frac{1}{70} + \frac{1}{150}$$

$$\frac{1}{f} = \frac{-150+70}{70\times150} = \frac{-80}{70\times150} = -\frac{80}{10500}$$

$$f = \frac{-1050}{8} = -131.25\,\text{cm}$$

As the final focal length is negative, the combination of two lenses is a diverging system of lenses.

The power of the combination is,

$$P = \frac{1}{f} = \frac{1}{-1.3125\,\text{m}} = -0.76\ \text{D}$$

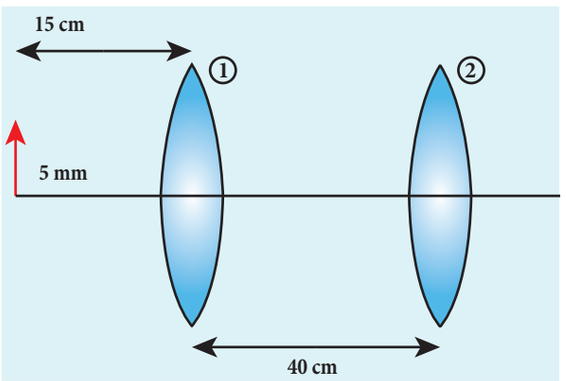
Points to Ponder





System of combination of lenses is commonly used in designing lenses for cameras, microscopes, telescopes and other optical instruments. They produce better magnification and sharpness of the image.

EXAMPLE 6.17



An object of 5 mm height is placed at a distance of 15 cm from a convex lens of focal length 10 cm. A second lens of focal length 5 cm is placed 40 cm from the first lens and 55 cm from the object. Find (a) the position of the final image, (b) its nature and (c) its size.

Solution

Given, $h = 5$ mm $= 0.5$ cm, $u_1 = -15$ cm, $f_1 = 10$ cm, $f_2 = 5$ cm, $d = 40$ cm

For the first lens, the lens equation is,

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

Substituting the values,

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